FORMULA MAP · START WITH THE ENERGY STORY

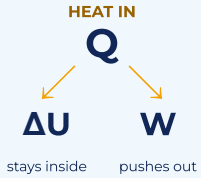
Heat in. *Energy kept.* Work out.

One law connects everything. Find what stays fixed, then fill in the zeros.

THE FIRST LAW

$$Q = \Delta U + W$$

Heat received = energy stored + work done by the gas.



Keep the signs

Heat enters: $Q > 0$
Expansion: $W > 0$
Compression: $W < 0$

THE GAS LAW

$$PV = nRT$$

ENERGY STORED

$$U = \frac{f}{2}nRT = \frac{f}{2}PV$$

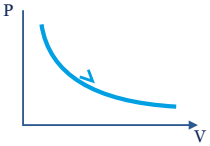
ANY IDEAL-GAS PROCESS

$$\Delta U = nC_v\Delta T$$

Same endpoints → same ΔU.

T Isothermal

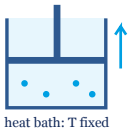
T stays fixed



$$\Delta U = 0 \quad Q = W = nRT\ln\frac{V_2}{V_1}$$

$PV = \text{constant}$ · $W = nRT\ln(P_1/P_2)$

Same temperature → same U. Heat becomes work. $C \to \infty$ when heat flows.



P Isobaric

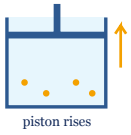
P stays fixed



$$W = P\Delta V = nR\Delta T \qquad Q = nC_p\Delta T$$

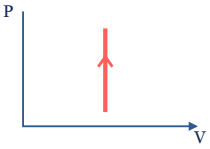
$V/T = \text{constant}$ · $C = C_p$

Free piston → heat warms AND pushes. $Q : \Delta U : W = C_p : C_v : R$



V Isochoric

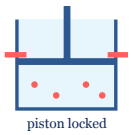
V stays fixed



$$W = 0 \qquad Q = \Delta U = nC_v\Delta T$$

$P/T = \text{constant}$ · $C = C_v$

Locked piston → no movement → no work. All the heat stays inside.



◆ Heat capacity shortcuts

$$C_v = \frac{f}{2}R \quad C_p = C_v + R \quad \gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f}$$

$$C_v = \frac{R}{\gamma - 1} \qquad C_p = \frac{\gamma R}{\gamma - 1}$$

Gas	f	C_v/R	C_p/R	γ
One atom	3	3/2	5/2	5/3
Two atoms	5	5/2	7/2	7/5
Nonlinear rigid	6	3	4	4/3

$C = \delta Q/(n\,dT)$; for constant C, $Q = nC\Delta T$.

◆ Mixtures: each mole gets a vote

$$C_{v,\text{mix}} = \frac{n_1C_{v1} + n_2C_{v2}}{n_1 + n_2}$$

$$f_{\text{mix}} = \frac{n_1f_1 + n_2f_2}{n_1 + n_2} \qquad \gamma_{\text{mix}} = 1 + \frac{2}{f_{\text{mix}}}$$

$C_{p,\text{mix}} = C_{v,\text{mix}} + R$. **Never average γ.**
Equal moles, f = 3 and 5 → $f_{\text{mix}} = 4$, $\gamma_{\text{mix}} = 3/2$.

At constant P: $\Delta U/Q = 1/\gamma$; $W/Q = (\gamma - 1)/\gamma$.

Water, metal & phase change

$Q = mc\Delta T$ or mL; $W = P\Delta V$ (P fixed)
$V = m/\rho$; $\Delta U = Q - W$. Heat-input rate − work-output rate = dU/dt.

Read the symbols

n = moles; f = degrees of freedom; C is per mole.
1 = initial, 2 = final; Δ = final − initial.
U is a state function; Q and W depend on the path.

ADIABATIC CHANGES · FREE EXPANSION · CYCLES

Spend stored energy. *Close the loop.*

No heat? The gas uses its own energy. Back to the start? The energy change is zero.

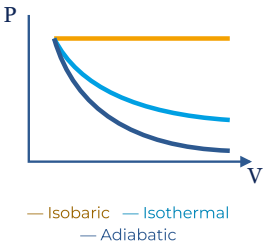
Q Adiabatic: no heat in or out

Q = 0

$$W = -\Delta U = nC_v(T_1 - T_2)$$

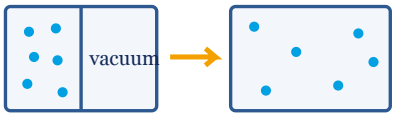
$$W = \frac{nR(T_1 - T_2)}{\gamma - 1} = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$$

$PV^\gamma = \text{const}$ → $TV^{\gamma-1} = \text{const}$ → $P \propto T^{\gamma/(\gamma-1)}$



Expansion does work → T falls. Compression → T rises.
$C = 0$. The three relations above require a reversible adiabatic path.

Same initial state + same final V: expansion $W_P > W_T > W_{ad}$. Compression: reverse the order for $|W|$.
At the same P,V: slopes are $-P/V$ (isothermal) and $-\gamma P/V$ (adiabatic).



Remove the partition. Nothing gets pushed.

Free expansion: no one to push

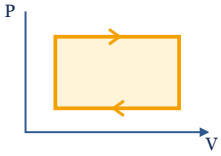
$$Q = W = \Delta U = 0 \quad \Delta T = 0$$

Insulated box + vacuum + ideal gas. $P_1V_1 = P_2V_2$ at the endpoints.
No equilibrium P–V path. Do not use $PV^\gamma = \text{const}$.

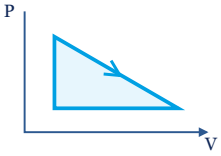
THERMODYNAMIC CYCLES · BACK TO THE SAME STATE

$$\Delta U_{\text{cycle}} = 0 \rightarrow Q_{\text{net}} = W_{\text{net}} = \oint P\,dV$$

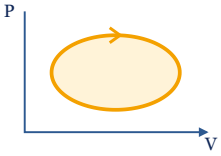
Net work = signed area inside the loop. Clockwise is positive when P is vertical and V is horizontal.



Rectangle
$|W| = \Delta P\,\Delta V$



Triangle
$|W| = \frac{1}{2}\Delta P\,\Delta V$



Circle / ellipse
$|W| = \frac{\pi}{4}\Delta P\,\Delta V$

Read direction first

Clockwise: $+W$
Anticlockwise: $-W$

ΔP and ΔV are **full spans**.
Swap axes → reverse signs.
Several loops → add signed areas.

◆ Long cycle? Make a table.

Name each step → fill the zeros → $Q = \Delta U + W$.

Step	Q / J	ΔU / J	W / J
AB · T fixed	+700	0	+700
BC · Q = 0	0	−400	+400
CA · return	−100	+400	−500
Cycle	**+600**	**0**	**+600**

Check: $\sum \Delta U = 0$; $\sum Q = \sum W$.
CA: 500 J of work is done **on** the gas.

◆ Heat in is not net heat

Q_{in} = sum of positive Q's.
Q_{out} = magnitude of negative Q's added together.

$$Q_{\text{net}} = Q_{\text{in}} - Q_{\text{out}} = W_{\text{net}}$$

Cycle efficiency $\eta = \frac{W_{\text{net}}}{Q_{\text{in}}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$

Power = work per cycle × cycles per second.
N cycles/min → $\text{Power} = NW_{\text{net}}/60$.

Other paths

$W = \int P\,dV$; straight line: $W = \frac{1}{2}(P_1 + P_2)(V_2 - V_1)$.
Curved P(V)? Integrate. P–T area is not work.
P–T: vertical = T fixed; horizontal = P fixed.
$W_{\text{on}} = -W_{\text{by}}$.

Units save marks

Pa·m³ = J; kPa·L = J; kPa·cm³ = 10^{-3} J.
L·atm ≈ 101.3 J; R = 8.31 J mol^{-1} K^{-1}.
Use kelvin: T(K) = t(°C) + 273.15.

Ideal gas, fixed n and constant heat capacities. Gas-pressure work paths assume reversible processes. Zeroth law: $T_A = T_C$, $T_B = T_C \Rightarrow T_A = T_B$.

POLYTROPIC PROCESSES · HEAT ENGINES · EFFICIENCY

One exponent. *One engine.*

Find the exponent to choose the process. Count only heat taken in to find efficiency.

A FAMILY OF GAS PROCESSES

$$PV^x = K$$

x is the **process exponent**, not the number of moles n.
These relations describe a quasistatic ideal-gas path with constant x.

x	Process	Molar C
0	Isobaric	C_p
1	Isothermal	∞^*
γ	Reversible adiabatic	0
$x \to \infty$	Isochoric limit	C_v

◆ Work: put x in place of γ

$$W = \frac{P_1V_1 - P_2V_2}{x-1} = \frac{nR\Delta T}{1-x}$$

Use only for x ≠ 1. At x = 1, use $W = nRT\ln(V_2/V_1)$ instead.

◆ Heat capacity along the path

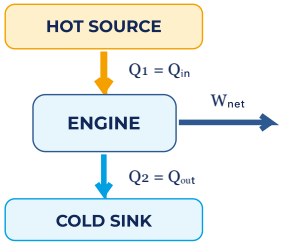
$$C = C_v + \frac{R}{1-x} \qquad Q = nC\Delta T$$

For x ≠ 1; $\Delta U = nC_v\Delta T$ still works.
Negative C: $1 < x < \gamma$ (for $\gamma > 1$).

Decode the question

$P \propto V \Rightarrow x = -1$ | $P^2V = K \Rightarrow x = \frac{1}{2}$ | $PT = K \Rightarrow x = \frac{1}{2}$ | $V \propto T^2 \Rightarrow x = \frac{1}{2}$

If T appears, substitute $T = PV/(nR)$. Then make the power of P equal to 1. *Isothermal: heat flows while T stays fixed.



Heat in = useful work + heat rejected

$$Q_1 = W_{\text{net}} + Q_2$$

$Q_1 = Q_{\text{in}} > 0$: sum of **positive** Q's.
$Q_2 = Q_{\text{out}} > 0$: sum of the **magnitudes of negative** Q's.

$$W_{\text{net}} = Q_1 - Q_2$$

The working substance completes a cycle, so $\Delta U = 0$.

EFFICIENCY USES HEAT ABSORBED

$$\eta = \frac{W_{\text{net}}}{\text{heat absorbed}} = \frac{W_{\text{net}}}{Q_{\text{in}}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$$

Useful output ÷ energy supplied. Percentage efficiency = $100\eta\%$.

Common mistake: putting net heat in the denominator. $W_{\text{net}}/Q_{\text{net}} = 1$ for a cycle; that is **not** engine efficiency.

◆ Work straight from a P–V graph

$$nR\Delta T = \Delta(PV)$$

$$\Delta U = \frac{C_v}{R}\Delta(PV)$$

Constant V: $Q = (C_v/R)\Delta(PV)$.
Constant P: $Q = (C_p/R)\Delta(PV)$.



One-atom gas, clockwise rectangle:
P runs from P_0 to $2P_0$.
V runs from V_0 to $2V_0$.
Vertical heating + top expansion take heat in.

$W = P_0V_0$; $Q_{\text{in}} = (\frac{3}{2}+5)P_0V_0$.

$$\eta = \frac{P_0V_0}{(13/2)P_0V_0} = \frac{2}{13} \approx 15.4\%$$

◆ Count the positive Q's only

For the cycle table on page 2:
$Q = +700,\ 0,\ -100$ J.

$$Q_{\text{in}} = 700\text{ J}, \quad Q_{\text{out}} = 100\text{ J}$$

$$W_{\text{net}} = 600\text{ J} \quad \Rightarrow \quad \eta = \frac{600}{700} = 85.7\%$$

Another quick check: 1000 J absorbed, 700 J rejected → W = 300 J, η = 30%.

Power: repeat the work each second

$$\text{Power} = W_{\text{net}} \times \text{cycles per second}$$

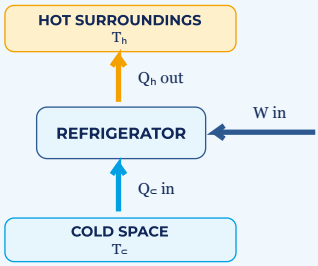
N cycles per minute → Power $= NW_{\text{net}}/60$.
2000 J in, 1500 J out, 10 cycles/s:
η = 25%; power = 5000 W.

Ideal gas; fixed amount and constant heat capacities. Q_1 and Q_2 are positive heat magnitudes, unlike signed Q for an individual step. Formula-sheet examples assume only P–V work.

REFRIGERATORS · HEAT PUMPS · COEFFICIENT OF PERFORMANCE

Use work. *Move heat uphill.*

A refrigerator removes heat from a cold space. Electrical work helps send it to warmer surroundings.



THE ENERGY BALANCE STILL WORKS

$$Q_h = Q_c + W_{\text{in}}$$

$$W_{\text{in}} = Q_h - Q_c$$

Q_c: heat removed from the cold space.
Q_h: heat delivered to the warmer surroundings.
$W_{\text{in}} > 0$: work supplied to the device.

Sign reminder: over the cycle, $\Delta U = 0$.
Earlier pages use work **by** the gas: $W_{\text{by}} = -W_{\text{in}}$.

◆ Refrigerator / air conditioner

Useful result: remove heat from the cold space.

$$\beta = \text{COP}_{\text{R}} = \frac{Q_c}{W_{\text{in}}} = \frac{Q_c}{Q_h - Q_c}$$

β = 4 means 4 J of cooling for every 1 J of work.
COP can exceed 1. It is not thermal efficiency.

◆ Heat pump

Useful result: deliver heat to the warm space.

$$\text{COP}_{\text{HP}} = \frac{Q_h}{W_{\text{in}}} = \text{COP}_{\text{R}} + 1$$

Same device and same cycle, different useful output.
Remember: hot-side heat = cold-side heat + work.

K Carnot limit: the best possible COP

Temperatures in kelvin

Two fixed-temperature reservoirs, $T_h > T_c$. Equality below requires a **reversible** cycle.

REFRIGERATOR

$$\beta_{\max} = \frac{T_c}{T_h - T_c}$$

HEAT PUMP

$$\text{COP}_{\text{HP,max}} = \frac{T_h}{T_h - T_c}$$

MINIMUM WORK FOR Q_c

$$W_{\min} = Q_c\left(\frac{T_h}{T_c} - 1\right)$$

Reversible: $Q_c/Q_h = T_c/T_h$. Actual $\text{COP} \le \text{COP}_{\text{Carnot}}$.
Smaller temperature gap → larger ideal COP. For a Carnot engine between the same temperatures:
$\eta_C = 1 - T_c/T_h$, $\beta_{\max} = (1-\eta_C)/\eta_C$, $\text{COP}_{\text{HP,max}} = 1/\eta_C$.

◆ From energy to power

$$\dot{Q}_c = \beta P_{\text{in}} \qquad \dot{Q}_h = (\beta + 1)P_{\text{in}}$$

$$W_{\text{in}} = P_{\text{in}} t \qquad t = \frac{Q_c}{\beta P_{\text{in}}}$$

Dots mean energy per second. Use watts and seconds.
Time formula assumes constant COP and input power.

◆ Find the cooling load first

$$Q_c = mc(T_i - T_f) \quad (T_i > T_f)$$

$$Q_c = mL \quad \text{during freezing}$$

Cooling + freezing? Add the heat removed in each stage.
Then $W_{\text{in}} = Q_c/\beta$. Match c and L to the material.

◆ Example: count the joules

A refrigerator removes 1200 J and uses 300 J of work.

$$\beta = \frac{1200}{300} = 4 \qquad Q_h = 1200 + 300 = 1500\ \text{J}$$

$$\text{COP}_{\text{HP}} = \frac{1500}{300} = 5 = 4 + 1$$

Rate check: β = 3, input power = 200 W.
Cooling = 600 W; heat delivered = 800 W.
Removing 36 kJ takes $36000/600 = 60\,\text{s}$.

◆ Example: ideal temperature limit

Carnot refrigerator: $T_c = 270\,\text{K}$, $T_h = 300\,\text{K}$.

$$\beta_{\max} = \frac{270}{300 - 270} = 9$$

$$Q_c = 900\,\text{J} \quad \Rightarrow \quad W_{\min} = \frac{900}{9} = 100\,\text{J}$$

$Q_h = 1000\,\text{J}$; ideal heat-pump COP = 10.
Temperatures alone give the **ideal limit**, not the COP of a real refrigerator.

Choose the numerator: engine → work out; refrigerator → heat removed from cold space; heat pump → heat delivered to warm space. **For both cooling devices, the denominator is work input.**

All Q and W symbols on this page are positive magnitudes. Reference checks: OpenStax, University Physics Vol. 2, §4.3 and §4.5. Diagrams and worked examples drawn for this sheet.